Tutorial on Discrete Diffusion

Forward Noising, Backward Couplings, Learning, and Operator Matching

1 Forward Noising Process

We work on a finite state space $\mathcal{X} = \{1, \dots, n\}$. Let

$$h^t \in \mathbb{R}_+^n, \qquad \sum_{i=1}^n h_i^t = 1$$

be the histogram (marginal law) at time t.

The forward kernel is row-stochastic:

$$P \in \mathbb{R}_+^{n \times n}, \qquad P_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i), \qquad P\mathbf{1} = \mathbf{1}.$$

Forward marginals follow

$$h^{t+1} = P^\top h^t, \tag{1}$$

for $t = 0, \dots, T-1$, hence we have $T+1$ histograms $(h^0, \dots, h^T)$.

Examples of forward kernels (detailed).

- **Local geometric kernel.** For interior states, mass is split on neighbors and self:

  $$P_{ij} = \frac{1}{3} \mathbf{1}_{j \in \{i-1, i, i+1\}},$$

  with boundary rows set by a chosen convention (reflecting or periodic). This kernel encodes local motion and typically smooths histograms spatially.

- **Gaussian convolution kernel.** For a bandwidth $\sigma > 0$,

  $$P_{ij} \propto \exp\left(-\frac{(i-j)^2}{2\sigma^2}\right), \qquad \sum_j P_{ij} = 1.$$

  Large σ gives stronger global mixing; small σ remains local.

- **Masking/absorbing kernel.** Choose a special absorbing state n and set

  $$P_{ii} = 1 - \varepsilon, \quad P_{in} = \varepsilon \ (i \neq n), \quad P_{nn} = 1.$$

  The process progressively transfers mass toward state n.

Forward particle process. Define one Markov process $(X^t)_{t=0}^T$ by

$$X^0 \sim h^0, \qquad X^{t+1} \sim P\big(X^t, \cdot\big).$$

Then

$$h^t = \mathrm{Law}(X^t),$$

and (1) is exactly the evolution of this law.

Forward figures. The script `figures/figures.py` generates forward visualizations using the same Gaussian forward kernel as the notebook,

$$P_{ij} \propto \exp\left(-\frac{(i-j)^2}{2\sigma^2}\right), \qquad \sigma = 0.75.$$

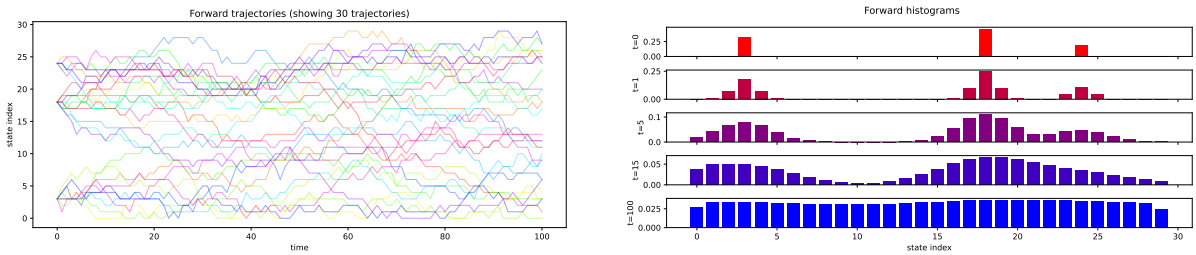

Figure 1: Forward process (left: trajectories, right: vertically stacked histogram snapshots) generated with the Gaussian kernel P.

2 Backward Kernels and Coupling

2.1 Admissible kernels between consecutive marginals

Fix $a, b \in \Delta_n$ (probability simplex). Define

$$\mathcal{T}(a,b) = \left\{K \in \mathbb{R}_+^{n\times n} : \; K\mathbf{1} = \mathbf{1}, \; K^\top a = b\right\}.$$

Thus $P \in \mathcal{T}(h^t, h^{t+1})$ by (1).

An inverse (backward) kernel at step t is any

$$Q^t \in \mathcal{T}(h^{t+1}, h^t),$$

that is,

$$Q^t\mathbf{1} = \mathbf{1}, \qquad (Q^t)^\top h^{t+1} = h^t.$$

So there are exactly T backward kernels $Q^0, \ldots, Q^{T-1}$ connecting $t+1 \to t$.

2.2 Backward particle process

Given admissible kernels $(Q^t)_{t=0}^{T-1}$, define $(Y^t)_{t=0}^{T}$ backward by

$$Y^T \sim h^T, \qquad Y^t \sim Q^t(Y^{t+1}, \cdot), \quad t = T-1, \ldots, 0.$$

If each $Q^t \in \mathcal{T}(h^{t+1}, h^t)$, then

$$\mathrm{Law}(Y^t) = h^t, \qquad t = 0, \ldots, T,$$

so backward recursion terminates at h^0.

2.3 Bayes inverse as a canonical admissible reverse kernel

Definition (Bayes kernel). Given a forward chain with kernel P and marginals (h^t), the Bayes kernel at time t is

$$Q_{ij}^{\mathrm{B},t} := \mathbb{P}(X^t = j \mid X^{t+1} = i).$$

Proposition 1. *Let $a = h^t$, $b = h^{t+1}$ and $P \in \mathcal{T}(a, b)$. Then for $b_i > 0$,*

$$Q^{\mathrm{B}}_{ij} = \frac{a_j P_{ji}}{b_i}. \tag{2}$$

In matrix form,

$$Q^{\mathrm{B}} = \operatorname{diag}(b)^{+} P^{\top} \operatorname{diag}(a). \tag{3}$$

Moreover, $Q^{\mathrm{B}} \in \mathcal{T}(b, a)$, i.e. it is an admissible inverse kernel.

Bayesian inversion map. Define

$$\mathcal{B}_{a,b} : \mathcal{T}(a, b) \to \mathcal{T}(b, a), \qquad \mathcal{B}_{a,b}(K) := \operatorname{diag}(b)^{+} K^{\top} \operatorname{diag}(a).$$

On supports where $a_i, b_i > 0$, this map is a bijection, with inverse

$$\mathcal{B}^{-1}_{a,b}(L) = \mathcal{B}_{b,a}(L) = \operatorname{diag}(a)^{+} L^{\top} \operatorname{diag}(b).$$

Therefore Bayesian inversion is not only a way to build one reverse kernel from P; it defines a one-to-one correspondence between admissible forward kernels and admissible backward kernels.

When $b_i = 0$, row i does not affect $(Q^{\mathrm{B}})^{\top} b$ and can be set by pseudo-inverse convention plus row renormalization.

Backward figures. Using $Q^t = \mathcal{B}_{h^t, h^{t+1}}(P)$ at each step, the backward process

$$Y^T \sim h^T, \qquad Y^t \sim Q^t(Y^{t+1}, \cdot)$$

has marginals h^t for $t = T, \ldots, 0$. Figure 2 displays backward trajectories (left) and empirical histograms of Y^t (right), generated with the Bayes inverse kernels associated with the same Gaussian forward kernel P.

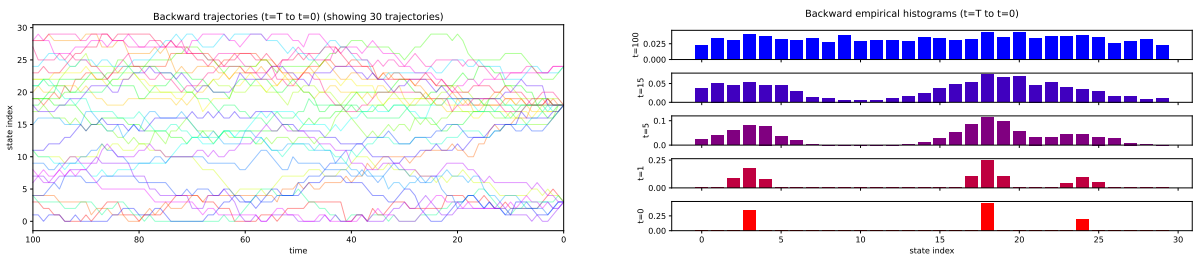

Figure 2: Backward process (left: trajectories in backward time, right: vertically stacked empirical histogram snapshots of Y_t). Kernel: Bayes inverse of the Gaussian forward kernel P.

2.4 Coupling polytope and bijection with kernels

Define

$$\Pi(a, b) = \left\{ \pi \in \mathbb{R}^{n \times n}_{+} : \ \pi \mathbf{1} = a, \ \pi^{\top} \mathbf{1} = b \right\}.$$

Kernel–coupling bijection (on $\operatorname{supp}(a)$):

$$\Phi_a : \mathcal{T}(a, b) \to \Pi(a, b), \qquad \Phi_a(K) = \operatorname{diag}(a) K,$$

with inverse

$$\Phi_a^{-1}(\pi) = \operatorname{diag}(a)^{+} \pi,$$

where $\operatorname{diag}(a)^{+}$ is the diagonal pseudo-inverse ($1/a_i$ if $a_i > 0$, else 0).

Hence choosing a Markov kernel in $\mathcal{T}(a, b)$ is equivalent to choosing a coupling in $\Pi(a, b)$.

Interleaving with transposition and Bayes kernel as transposed coupling. Let

$$\mathsf{T} : \Pi(a,b) \to \Pi(b,a), \qquad \mathsf{T}(\pi) = \pi^\top.$$

Define similarly

$$\Phi_b : \mathcal{T}(b,a) \to \Pi(b,a), \qquad \Phi_b(L) = \mathrm{diag}(b)L.$$

Then the maps commute:

$$\mathsf{T} \circ \Phi_a = \Phi_b \circ \mathcal{B}_{a,b},$$

equivalently

$$\mathcal{B}_{a,b} = \Phi_b^{-1} \circ \mathsf{T} \circ \Phi_a.$$

So Bayesian inversion in kernel coordinates is exactly transposition in coupling coordinates. In particular, for $P \in \mathcal{T}(a,b)$ with $\pi = \Phi_a(P) = \mathrm{diag}(a)P$, the Bayes kernel is

$$Q^{\mathrm{B}} = \Phi_b^{-1}\big(\mathsf{T}(\pi)\big) = \mathrm{diag}(b)^+\pi^\top.$$

This is summarized by the commutative diagram (with arrows in both directions):

$$\begin{tikzcd}
\mathcal{T}(a,b) \arrow[r, shift left, "\Phi_a"] \arrow[d, shift left, "\mathcal{B}_{a,b}"] & \Pi(a,b) \arrow[l, shift left, "\Phi_a^{-1}"] \arrow[d, leftrightarrow, "\mathsf{T}"] \\
\mathcal{T}(b,a) \arrow[r, shift left, "\Phi_b^{-1}"] \arrow[u, shift left, "\mathcal{B}_{b,a}"] & \Pi(b,a) \arrow[l, shift left, "\Phi_b"]
\end{tikzcd}$$

3 Learning the Reverse Kernel

3.1 Statistical formulation

Let $Q_\theta^t(i,j)$ be a neural network parameterization such that, for each (t,i),

$$Q_\theta^t(i,\cdot) \in \Delta_n$$

(typically enforced by a softmax over j).

Assume we observe i.i.d. samples

$$\big(X_k^{t_k}, X_k^{t_k+1}, t_k\big)_{k=1}^N,$$

where t_k are i.i.d. in $\{0,\dots,T-1\}$ (e.g. uniform), and

$$X_k^{t_k+1} \sim P\Big(X_k^{t_k}, \cdot\Big), \qquad X_k^{t_k} \sim h^{t_k}.$$

Equivalently, each triplet is drawn from the joint law induced by the forward noising process:

$$t \sim \pi \text{ on } \{0,\dots,T-1\}, \quad X^t \sim h^t, \quad X^{t+1} \mid X^t \sim P(X^t,\cdot),$$

then set $(t_k, X_k^{t_k}, X_k^{t_k+1})$ as i.i.d. copies of (t, X^t, X^{t+1}). The empirical negative log-likelihood is

$$L_N(\theta) := \frac{1}{N}\sum_{k=1}^N \Big[-\log Q_\theta^{t_k}\big(X_k^{t_k+1}, X_k^{t_k}\big)\Big]. \tag{4}$$

Let θ_N be a global minimizer of L_N.

3.2 Population target and consistency interpretation

Define the Bayes reverse kernels

$$Q_\star^t(i,j) := \mathbb{P}(X^t = j \mid X^{t+1} = i,\, t),$$

which coincide with the canonical Bayes inverse from Section 2. The population risk is

$$L(\theta) = \mathbb{E}_{\substack{t\sim\pi \\ X^t\sim h^t \\ X^{t+1}|X^t\sim P(X^t,\cdot)}} \left[-\log Q_\theta^t(X^{t+1}, X^t)\right].$$

Proposition 2 (Population NLL decomposition)**.** *Let $Q_\theta^t(i,\cdot) \in \Delta_n$ for all (t,i). Then*

$$L(\theta) = C + \sum_{t=0}^{T-1} \pi(t) \sum_{i=1}^{n} h_i^{t+1}\, \mathrm{KL}\big(Q_\star^t(i,\cdot) \,\|\, Q_\theta^t(i,\cdot)\big), \tag{5}$$

where

$$C = \sum_{t=0}^{T-1} \pi(t) \sum_{i=1}^{n} h_i^{t+1}\, H\big(Q_\star^t(i,\cdot)\big)$$

is independent of θ.

Proof. Condition on $(t, X^{t+1} = i)$ and use $X^t \mid (t, X^{t+1} = i) \sim Q_\star^t(i,\cdot)$:

$$L(\theta) = \sum_t \pi(t) \sum_i h_i^{t+1} \sum_j Q_\star^t(i,j)\big[-\log Q_\theta^t(i,j)\big].$$

Add and subtract $\log Q_\star^t(i,j)$ inside the sum:

$$\sum_j Q_\star^t(i,j)\big[-\log Q_\theta^t(i,j)\big] = H(Q_\star^t(i,\cdot)) + \mathrm{KL}(Q_\star^t(i,\cdot)\|Q_\theta^t(i,\cdot)).$$

Summing over (t,i) gives (5). □

Therefore, minimizing $L(\theta)$ is equivalent to minimizing

$$\sum_{t=0}^{T-1} \pi(t) \sum_{i=1}^{n} h_i^{t+1}\, \mathrm{KL}\big(Q_\star^t(i,\cdot) \,\|\, Q_\theta^t(i,\cdot)\big).$$

Proposition 3 (Consistency, informal)**.** *Assume:*

- *the model class is rich enough to represent $Q_\star$ (well-specified/universal approximation),*
- *empirical risk minimization is statistically consistent (uniform LLN / Glivenko–Cantelli type conditions),*
- *optimization finds a global minimizer (or asymptotically equivalent minimizers).*

Then any limit point of Q_{θ_N} minimizes L and agrees with $Q_\star^t(i,\cdot)$ for all (t,i) such that $\pi(t)\, h_i^{t+1} > 0$. Hence the limit kernel is admissible:

$$Q_{\theta^\star}^t \in \mathcal{T}(h^{t+1}, h^t), \qquad t = 0, \ldots, T-1.$$

Important practical caveat. For finite N, minimizing (4) enforces row-stochasticity (via softmax) but does *not* guarantee exact marginal constraints

$$(Q_\theta^t)^\top h^{t+1} = h^t.$$

Exact admissibility is an asymptotic/population property under the assumptions above, or can be encouraged by explicit constraints/penalties.

3.3 Neural parameterization example

For 1D geometric histograms, a simple input encoding is

$$\text{position}(i) \parallel (t/T),$$

with $\text{position}(i) = i/(n-1)$ (or i/n). If no spatial prior is available, one may use $\text{position}(i) = \text{onehot}(i)$. A practical architecture is a 2-layer MLP with one ReLU and a final softmax over j [1, 3, 2, 4].

4 Connection with Classical Gaussian Diffusion

4.1 Additive Gaussian forward noising

In continuous state space ($x \in \mathbb{R}^d$), a standard forward step is

$$X^{t+\tau} = X^t + \sqrt{2\tau}\,\xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

so

$$p_\tau(x^{t+\tau} \mid x^t) = \mathcal{N}(x^t,\, 2\tau I_d).$$

More generally, for an SDE

$$dX^t = f(X^t, t)\,dt + g(t)\,dW_t,$$

the short-time kernel is Gaussian with mean $x + \tau f(x,t)$ and covariance $\tau g(t)^2 I_d$.

4.2 Bayes reverse conditional and score expansion

Let p_t be the density of X^t. By Bayes rule,

$$p(x^t \mid x^{t+\tau}) \propto p_\tau(x^{t+\tau} \mid x^t)\, p_t(x^t).$$

For small τ, a first-order expansion yields a Gaussian approximation

$$X^t \mid X^{t+\tau} = y \approx \mathcal{N}\Big(y - \tau f(y,t) + \tau g(t)^2 \nabla_y \log p_t(y),\ \tau g(t)^2 I_d\Big),$$

where the score is $s_t(y) = \nabla_y \log p_t(y)$.

4.3 Limit $\tau \to 0$: reverse-time SDE

Thus sampling from the reverse conditionals converges to the reverse-time SDE

$$dX^t = \big(f(X^t,t) - g(t)^2 \nabla \log p_t(X^t)\big)\,dt + g(t)\,d\bar{W}_t$$

(time run backward). For pure additive Gaussian noising ($f = 0$, $g = \sqrt{2}$):

$$dX^t = -2\,\nabla \log p_t(X^t)\,dt + \sqrt{2}\,d\bar{W}_t,$$

which is the classical score-based reverse diffusion dynamics.

5 Fine-Grid and Small-Step Limit: Denoising Form

We isolate here the 1D fine-grid limit and write the reverse-kernel parameterization directly in terms of a score-like field.

Let

$$x_i = i\Delta x, \qquad i = 1, \dots, n, \qquad \Delta x \to 0 \ (n \to \infty),$$

and consider the simple Gaussian forward step (no drift parameter):

$$P_{ij} \propto \exp\left(-\frac{(x_j - x_i)^2}{4\tau}\right), \qquad \Delta t = \tau.$$

In continuous notation this corresponds to

$$X^{t+\tau} \mid X^t = x \sim \mathcal{N}(x, 2\tau).$$

Denote by x' the state at time $t + \tau$ and by x the state at time t. The Bayes reverse conditional is

$$q_\star^\tau(x \mid x', t) \propto \exp\left(-\frac{(x' - x)^2}{4\tau}\right) p_t(x).$$

Write $s_t(x') \coloneqq \partial_x \log p_t(x')$. A first-order Taylor expansion

$$\log p_t(x) = \log p_t(x') + (x - x')\, s_t(x') + O((x - x')^2)$$

in the $\tau \to 0$ regime ($x - x' = O(\sqrt{\tau})$) gives, after completing the square,

$$q_\star^\tau(x \mid x', t) \approx \mathcal{N}\big(x' + 2\tau s_t(x'),\, 2\tau\big).$$

This gives the rationale for a reverse-kernel model driven by a field s_θ^t:

$$Q_\theta^t(x_j \mid x_i') \coloneqq \frac{\exp\left(-\frac{(x_j - x_i' - 2\tau s_\theta^t(x_i'))^2}{4\tau}\right)}{\sum_{\ell=1}^n \exp\left(-\frac{(x_\ell - x_i' - 2\tau s_\theta^t(x_i'))^2}{4\tau}\right)}. \tag{6}$$

So the discrete Bayes inverse kernel is expected to be locally approximated by this shifted-Gaussian family.

Proposition 4 (One-step NLL in denoising form, formal expansion)**.** *Assume the local approximation above and parameterization* (6)*. Then, up to* $o(1)$ *terms and additive constants independent of* θ*,*

$$L_\tau(\theta) = \mathbb{E}\big[-\log Q_\theta^t(X^t \mid X^{t+\tau})\big] = C_\tau + \frac{1}{4\tau}\,\mathbb{E}\Big[\big(X^t - X^{t+\tau} - 2\tau s_\theta^t(X^{t+\tau})\big)^2\Big].$$

Equivalently, defining the denoising target

$$D_\tau \coloneqq \frac{X^t - X^{t+\tau}}{2\tau},$$

minimizing L_τ *is equivalent (up to scaling/constants) to minimizing*

$$\mathbb{E}\Big[\big(D_\tau - s_\theta^t(X^{t+\tau})\big)^2\Big].$$

Proof sketch. Insert (6) into $-\log Q_\theta^t(X^t \mid X^{t+\tau})$. The log-normalizer contributes only through terms independent of X^t at leading order, absorbed in C_τ. Expanding the quadratic term yields exactly the stated squared-error form. □

Important clarification on the $\tau \to 0$ limit. A naive pointwise limit of

$$\mathbb{E}\Big[\big(D_\tau - s_\theta^t(X^{t+\tau})\big)^2\Big]$$

is not finite, because $D_\tau = (X^t - X^{t+\tau})/(2\tau)$ has variance of order $1/\tau$. Indeed, under $X^{t+\tau} = X^t + \sqrt{2\tau}\,\xi$,

$$D_\tau = -\frac{\xi}{\sqrt{2\tau}} + O(1),$$

so the unscaled loss diverges as $\tau \downarrow 0$.

The meaningful object is the *excess* risk (or equivalently the θ-dependent part), after removing the diverging noise floor:

$$L_\tau(\theta) - L_\tau(\theta_\star) \;\sim\; \tau\,\mathbb{E}\Big[\big(s_\theta^t(X^{t+\tau}) - s_t(X^{t+\tau})\big)^2\Big].$$

Hence one does not identify score matching from a single unscaled step; it appears after the proper small-step rescaling and, over a finite horizon, after summing/integrating many steps ($T \sim 1/\tau$):

$$\sum_{k=0}^{T-1}\big(L_\tau^{(k)}(\theta) - L_\tau^{(k)}(\theta_\star)\big) \;\longrightarrow\; \int_0^{\mathsf{T}} w(t)\,\mathbb{E}\Big[\big(s_\theta^t(X^t) - s^t(X^t)\big)^2\Big]\,dt$$

for an explicit weight $w(t)$ (equal to 1 in this normalized additive-noise setting). So your intuition is correct: the score-matching structure is fundamentally a multi-step/time-integrated limit, not a raw one-step $\tau \to 0$ limit of D_τ.

6 Continuous-Time Finite-Chain Formulation and Operator Matching

6.1 Continuous-time extension of the finite-state tutorial

Let $\mathcal{X} = \{1, \dots, n\}$ and let $(A_t)_{t\in[0,T]}$ be a time-dependent forward generator:

$$(A_t)_{ij} \geq 0 \;(j \neq i), \qquad (A_t)_{ii} = -\sum_{j\neq i}(A_t)_{ij}.$$

The forward marginal evolves by the Kolmogorov equation

$$\frac{d}{dt}h^t = A_t^\top h^t, \qquad h^0 \in \Delta_n.$$

For small step τ, the discrete kernel is

$$P^t = I + \tau A_t + o(\tau),$$

which recovers the tutorial recursion $h^{t+\tau} = (P^t)^\top h^t$.

6.2 Backward generator by infinitesimal Bayes inversion

Define reverse rates for $i \neq j$ by

$$(\widetilde{A}_t)_{ij} = \frac{h_j^t}{h_i^t}(A_t)_{ji}, \qquad (\widetilde{A}_t)_{ii} = -\sum_{j\neq i}(\widetilde{A}_t)_{ij}.$$

Then $Q^t = I + \tau\widetilde{A}_t + o(\tau)$ is the infinitesimal analogue of Bayes reverse kernels. Moreover, (h^t) is invariant for reverse-time dynamics in the sense

$$\frac{d}{dt}h^t = -\widetilde{A}_t^\top h^t.$$

This is the continuous-time counterpart of $Q^t \in \mathcal{T}(h^{t+\tau}, h^t)$.

6.3 Operator matching viewpoint in finite Markov chains

For test functions $\varphi : \mathcal{X} \to \mathbb{R}$, define forward/backward operators

$$(\mathcal{L}_t\varphi)(i) = \sum_j (A_t)_{ij}\big(\varphi(j) - \varphi(i)\big),$$

$$(\widetilde{\mathcal{L}}_t\varphi)(i) = \sum_j (\widetilde{A}_t)_{ij}\big(\varphi(j) - \varphi(i)\big).$$

The weak-form evolution is

$$\frac{d}{dt}\langle h^t, \varphi\rangle = \langle h^t, \mathcal{L}_t\varphi\rangle,$$

which is the finite-state analogue of the weak continuity-equation formulation used in operator flow matching.

To make operator matching identifiable, one must average over a rich family of test functions. Let $\Phi \subset \mathbb{R}^{\mathcal{X}}$ be a function class and let ρ be a sampling law on Φ. Given a parametric reverse generator $\widetilde{A}_\theta(t)$ (equivalently operator $\widetilde{\mathcal{L}}_\theta$), define

$$\mathcal{J}_{\mathrm{op}}(\theta) \coloneqq \mathbb{E}_t\, \mathbb{E}_{\varphi\sim\rho}\, \mathbb{E}_{i\sim h^t}\Big[\big((\widetilde{\mathcal{L}}_\theta)_t\varphi(i) - (\widetilde{\mathcal{L}}_\star)_t\varphi(i)\big)^2\Big],$$

where $\widetilde{\mathcal{L}}_\star$ is the Bayes reverse operator. In practice, ρ can be:

- uniform over a finite basis $\Phi = \{\varphi_m\}_{m=1}^M$ (e.g., canonical basis or low-frequency graph modes), or
- a random-feature distribution on functions.

Then

$$\mathcal{J}_{\mathrm{op}}(\theta) \approx \frac{1}{M}\sum_{m=1}^{M} \mathbb{E}_t\mathbb{E}_{i\sim h^t}\Big[\big((\widetilde{\mathcal{L}}_\theta)_t\varphi_m(i) - (\widetilde{\mathcal{L}}_\star)_t\varphi_m(i)\big)^2\Big].$$

This is the finite-chain specialization of matching transport operators/vector fields rather than directly matching full conditional distributions.

Link with kernel log-likelihood training. For small τ, minimizing transition NLL for kernels

$$Q_\theta^t = I + \tau\widetilde{A}_\theta(t) + o(\tau)$$

and denote off-diagonal reverse rates by

$$\widetilde{a}_{\theta,ij}(t) = (\widetilde{A}_\theta(t))_{ij}, \quad i \neq j, \qquad \widetilde{\lambda}_{\theta,i}(t) = \sum_{j\neq i} \widetilde{a}_{\theta,ij}(t).$$

For one small step, population NLL (cross-entropy) between $Q_\star^t$ and Q_θ^t expands as

$$\mathcal{L}_t(\theta) = C_t + \tau\sum_i h_i^t \sum_{j\neq i}\Big[\widetilde{a}_{\star,ij}\log\frac{\widetilde{a}_{\star,ij}}{\widetilde{a}_{\theta,ij}} - \widetilde{a}_{\star,ij} + \widetilde{a}_{\theta,ij}\Big] + o(\tau),$$

which is the weighted Poisson-KL between reverse rates. Near the optimum ($\widetilde{a}_{\theta,ij} \approx \widetilde{a}_{\star,ij}$),

$$\mathcal{L}_t(\theta) - \mathcal{L}_t(\star) = \frac{\tau}{2}\sum_i h_i^t \sum_{j\neq i}\frac{\big(\widetilde{a}_{\theta,ij} - \widetilde{a}_{\star,ij}\big)^2}{\widetilde{a}_{\star,ij}} + o\Big(\tau\|\widetilde{A}_\theta - \widetilde{A}_\star\|^2\Big),$$

which is the precise "quadratic local discrepancy between reverse rates".

Yes, this can be realized through a suitable choice of test-function distribution ρ. Take $\Phi = \{\mathbf{1}_{\{j\}}\}_{j=1}^n$ (canonical indicator functions), uniform ρ on Φ. Then

$$(\widetilde{\mathcal{L}}_t \mathbf{1}_{\{j\}})(i) = \begin{cases} \widetilde{a}_{ij}(t), & i \neq j, \\ -\widetilde{\lambda}_j(t), & i = j, \end{cases}$$

so $\mathcal{J}_{\mathrm{op}}$ becomes a weighted quadratic loss on off-diagonal rates plus escape rates. Hence, with this ρ, operator matching directly controls the same local objects as the small-τ NLL expansion. If Φ is dense/separating, $\mathcal{J}_{\mathrm{op}}(\theta) = 0$ implies $\widetilde{A}_\theta = \widetilde{A}_\star$ (on the support of h^t), exactly as in the Gaussian continuous-state limit where reverse-kernel NLL reduces to score regression.

References

[1] Jacob Austin, Daniel Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. *arXiv preprint arXiv:2107.03006*, 2021.

[2] Francis Bach and Saeed Saremi. Sampling binary data by denoising through score functions. *arXiv preprint arXiv:2502.00557*, 2025.

[3] Andrew Campbell, Joe Benton, Valentin De Bortoli, Tom Rainforth, George Deligiannidis, and Arnaud Doucet. A continuous time framework for discrete denoising models. *arXiv preprint arXiv:2205.14987*, 2022.

[4] Vincent Pauline, Tobias H"oppe, Kirill Neklyudov, Alexander Tong, Stefan Bauer, and Andrea Dittadi. Foundations of diffusion models in general state spaces: A self-contained introduction. *arXiv preprint arXiv:2512.05092*, 2025.